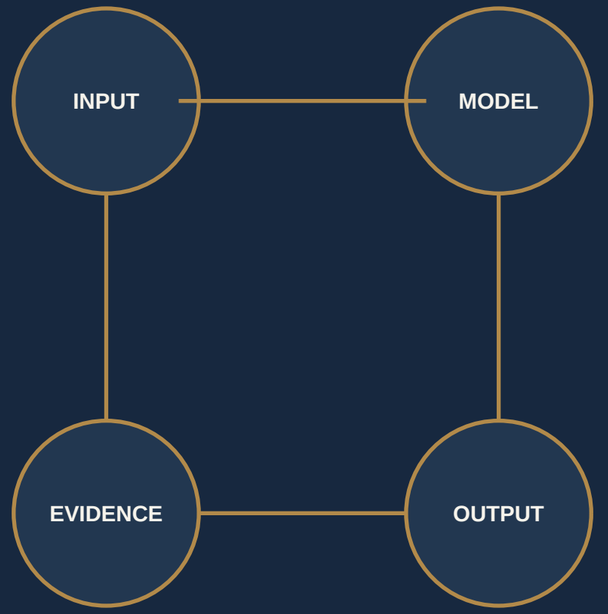
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

# Robotics Planning

AI.311 | University lecture series



# What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define state-space planning.
- Apply key methods including search.
- Build or analyze a practical system involving task planning.
- Evaluate assumptions, limitations, performance and responsible use.

# A connected learning journey

Each stage prepares the tools needed for the next.

01

**FOUNDATIONS**

State-space planning

02

**METHODS**

Search

03

**SYSTEMS**

Task planning

04

**PRACTICE**

Laboratory, evaluation and project

# The subject at a glance

A map of the ideas and techniques in this course

### 01 · State-space planning

State-space planning

### 02 · Search

Search

### 03 · Task planning

Task planning

### 04 · Motion planning

Motion planning

### 05 · Trajectory generation and ma

Trajectory generation and manipulation planning

### 06 · Mandatory practical: Implement

Mandatory practical: Implement planning algorithms and generate collision-free robot motions

# Core ideas, made explicit

AI.311 • Distinguish definitions, assumptions and evidence

- State-space planning
- Search
- Task planning
- Motion planning
- Trajectory generation and manipulation planning

**DEFINE → MODEL → TEST → INTERPRET**

# Core ideas, made explicit

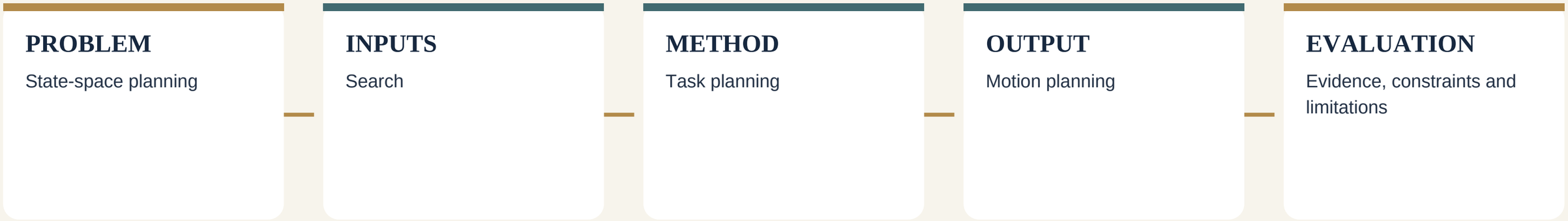
AI.311 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Implement planning algorithms and generate collision-free robot motions

**DEFINE → MODEL → TEST → INTERPRET**

# Follow the system through its stages

Use the process diagram to connect components and outcomes.




Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

# From concept to a real engineering decision

A short case study for discussion and worked-example narration

### SCENARIO

A real task involving state-space planning.

### CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

### DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

# Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using state-space planning.
- Implement or simulate a baseline using search.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

# Review · explain · apply

## KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

## PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

**Key takeaway: make the model visible, test it, and communicate its limits.**

Course-series slide design • detailed narration, demonstrations and assessments expand each module.